

ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \quad (1.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1.1b)$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \epsilon \frac{\partial E_x}{\partial t} + \sigma E_x \quad (1.2a)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \epsilon \frac{\partial E_y}{\partial t} + \sigma E_y \quad (1.2b)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \epsilon \frac{\partial E_z}{\partial t} + \sigma E_z \quad (1.2c)$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \quad (1.3a)$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \quad (1.3b)$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \quad (1.3c)$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i . The implementation uses

$\mathbf{E}_\alpha$: cell-centered in the α direction and nodal in the other two directions,

$\mathbf{H}_\alpha$: nodal in the α direction and cell-centered in the other two directions.

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i + \frac{1}{2}$. They are encoded in the **MultiFab** staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j + \frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j + 1$. Therefore

$$\left. \frac{\partial E_z}{\partial y} \right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}.$$

Similarly, $E_{x,i,j,k}$ is located at j , while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j - \frac{1}{2}$ and $j + \frac{1}{2}$, so

$$\left. \frac{\partial H_z}{\partial y} \right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}.$$

This is why the magnetic update below uses forward differences such as $j + 1$ and $k + 1$, while the electric update uses backward differences such as $j - 1$ and $k - 1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update **H** by a half step, then update **E** by a full step, and finally update **H** by another half step. Define the update coefficients

$$C_a = \frac{1 - \frac{\sigma \Delta t}{2\epsilon}}{1 + \frac{\sigma \Delta t}{2\epsilon}}, \quad C_b = \frac{\frac{\Delta t}{\epsilon}}{1 + \frac{\sigma \Delta t}{2\epsilon}}, \quad D_b = \frac{\Delta t}{\mu}, \quad (2.1)$$

where Δt is the time step, ϵ is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - \frac{D_b}{2} \left[\frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k+1}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta z} \right] \quad (2.2a)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - \frac{D_b}{2} \left[\frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} - \frac{E_{z,i+1,j,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta x} \right] \quad (2.2b)$$

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - \frac{D_b}{2} \left[\frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j+1,k}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta y} \right]. \quad (2.2c)$$

In the AMReX Yee grid, the magnetic field updates are implemented as

$$B_x(i, j, k) = \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.3a)$$

$$B_y(i, j, k) = \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.3b)$$

$$B_z(i, j, k) = \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.3c)$$

2.2 Electric Full-Step

$$E_{x,i+\frac{1}{2},j,k}^{n+1} = C_a E_{x,i+\frac{1}{2},j,k}^n + C_b \left[\frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.4a)$$

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_a E_{y,i,j+\frac{1}{2},k}^n + C_b \left[\frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.4b)$$

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_a E_{z,i,j,k+\frac{1}{2}}^n + C_b \left[\frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.4c)$$

In the AMReX Yee grid, the electric field updates are implemented as

$$E_x(i, j, k) + = c^2 \Delta t \left[\frac{B_z(i, j, k) - B_z(i, j - 1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k - 1)}{\Delta z} \right] \quad (2.5a)$$

$$E_y(i, j, k) + = c^2 \Delta t \left[\frac{B_x(i, j, k) - B_x(i, j, k - 1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i - 1, j, k)}{\Delta x} \right] \quad (2.5b)$$

$$E_z(i, j, k) + = c^2 \Delta t \left[\frac{B_y(i, j, k) - B_y(i - 1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j - 1, k)}{\Delta y} \right]. \quad (2.5c)$$

2.3 Second Magnetic Half-Step

$$H_{x, i, j + \frac{1}{2}, k + \frac{1}{2}}^{n+1} = H_{x, i, j + \frac{1}{2}, k + \frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{z, i, j+1, k+\frac{1}{2}}^{n+1} - E_{z, i, j, k+\frac{1}{2}}^{n+1}}{\Delta y} - \frac{E_{y, i, j+\frac{1}{2}, k+1}^{n+1} - E_{y, i, j+\frac{1}{2}, k}^{n+1}}{\Delta z} \right] \quad (2.6a)$$

$$H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^{n+1} = H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{x, i+\frac{1}{2}, j, k+1}^{n+1} - E_{x, i+\frac{1}{2}, j, k}^{n+1}}{\Delta z} - \frac{E_{z, i+1, j, k+\frac{1}{2}}^{n+1} - E_{z, i, j, k+\frac{1}{2}}^{n+1}}{\Delta x} \right] \quad (2.6b)$$

$$H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+1} = H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{y, i+1, j+\frac{1}{2}, k}^{n+1} - E_{y, i, j+\frac{1}{2}, k}^{n+1}}{\Delta x} - \frac{E_{x, i+\frac{1}{2}, j+1, k}^{n+1} - E_{x, i+\frac{1}{2}, j, k}^{n+1}}{\Delta y} \right]. \quad (2.6c)$$

In the AMReX Yee grid, the second magnetic field updates are implemented as

$$B_x(i, j, k) - = \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.7a)$$

$$B_y(i, j, k) - = \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.7b)$$

$$B_z(i, j, k) - = \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.7c)$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \quad c = \frac{1}{\sqrt{\mu\epsilon}}. \quad (2.8)$$

3 ADI Method

3.1 First ADI Half-Step ($n \rightarrow n + \frac{1}{2}$)

$$\left(\frac{\epsilon}{\Delta t} + \frac{1}{2}\sigma \right) E_x^{n+\frac{1}{2}} = \left(\frac{\epsilon}{\Delta t} - \frac{1}{2}\sigma \right) E_x^n + [H_z] \quad (3.1a)$$

$$E_{y, i, j+\frac{1}{2}, k}^{n+\frac{1}{2}} = C_a(i, j + \frac{1}{2}, k) \cdot E_{y, i, j+\frac{1}{2}, k}^n + C_b \cdot [H_z] \quad (3.1b)$$

Magnetic field updates:

$$H_{x, i, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x, i, j+\frac{1}{2}, k+\frac{1}{2}}^n + D_b \left(i, j + \frac{1}{2}, k + \frac{1}{2} \right) [\dots] \quad (3.2a)$$

$$H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} = H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^n + D_b \left(i + \frac{1}{2}, j + \frac{1}{2}, k \right) \left[\frac{E_{x, i+\frac{1}{2}, j+1, k}^{n+\frac{1}{2}} - E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}}{\Delta y} \right] \quad (3.2b)$$

Tridiagonal System formulation

Substitute (3.2b) into (3.1a) to eliminate $H_z^{n+\frac{1}{2}}$:

$$-\alpha_{x1}E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1}E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1}E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} = p_{x1}E_{x,i+\frac{1}{2},j,k}^n + \dots \quad (3.3a)$$

Where the coefficients are:

$$\begin{aligned} \alpha_{y1} &= D_b \left(i + \frac{1}{2}, j - \frac{1}{2}, k \right) / \Delta y^2 \\ \beta_{x1} &= 1/C_b \left(i + \frac{1}{2}, j, k \right) + \alpha_{y1} + \gamma_{y1} \\ \gamma_{x1} &= D_b \left(i + \frac{1}{2}, j + \frac{1}{2}, k \right) / \Delta y^2 \\ p_{x1} &= C_a \left(i + \frac{1}{2}, j, k \right) / C_b \left(i + \frac{1}{2}, j, k \right) \end{aligned}$$

Substitute (3.2a) into (3.1b):

$$-\alpha_{y1}E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}} + \beta_{y1}E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - \gamma_{y1}E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} = \dots \quad (3.3b)$$

With coefficient $\alpha_{z1} = D_b(i, j + \frac{1}{2}, k - \frac{1}{2}) / \Delta z^2$.

Substitute the corresponding magnetic update into the E_z equation:

$$-\alpha_{z1}E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \beta_{z1}E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \gamma_{z1}E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = \dots \quad (3.3c)$$

3. Second ADI Half-Step ($n + \frac{1}{2} \rightarrow n + 1$)

$$E_{x,i+\frac{1}{2},j,k}^{n+1} = C_a \left(i + \frac{1}{2}, j, k \right) \cdot E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \dots \quad (3.4a)$$

$$E_{y,i,j+1,k}^{n+1} = C_a(i, j + 1, k) \cdot E_{y,i,j+1,k}^{n+\frac{1}{2}} \dots \quad (3.4b)$$

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_a \left(i, j, k + \frac{1}{2} \right) \cdot E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \dots \quad (3.4c)$$

Magnetic field updates:

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + \dots \quad (3.5a)$$

Tridiagonal System formulation (Second Half-Step)

Substitute the corresponding magnetic update into (3.4b):

$$-\alpha_{y2}E_{y,i-1,j+\frac{1}{2},k}^{n+1} + \beta_{y2}E_{y,i,j+\frac{1}{2},k}^{n+1} - \gamma_{y2}E_{y,i+1,j+\frac{1}{2},k}^{n+1} = \dots \quad (3.6a)$$

Where:

$$\begin{aligned} \alpha_{y2} &= D_b \left(i - \frac{1}{2}, j + \frac{1}{2}, k \right) / \Delta x^2 \\ \beta_{y2} &= 1/C_b \left(i, j + \frac{1}{2}, k \right) + \alpha_{y2} + \gamma_{y2} \\ \gamma_{y2} &= D_b \left(i + \frac{1}{2}, j + \frac{1}{2}, k \right) / \Delta x^2 \end{aligned}$$

Substitute (3.5a) into (3.4c):

$$-\alpha_{z2}E_{z,i,j-1,k+\frac{1}{2}}^{n+1} + \beta_{z2}E_{z,i,j,k+\frac{1}{2}}^{n+1} - \gamma_{z2}E_{z,i,j+1,k+\frac{1}{2}}^{n+1} = \dots \quad (3.6b)$$

Where:

$$\alpha_{z2} = D_b \left(i, j - \frac{1}{2}, k + \frac{1}{2} \right) / \Delta y^2$$